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Chapter 12

Mineral Nutrition

Learning Objectives

The learner will be able to,

- *Recognise the need of mineral nutrition.*
- *Analyse the classification and criteria for essential minerals.*
- *Learn the techniques of Hydroponics and Aeroponics.*
- *Correlate different types of special modes of nutrition.*
- *Ability to recall and analyse nitrogen fixation.*

Chapter Outline

12.1 Classification of Minerals

12.2 Functions, mode of absorption, deficiency symptoms of Macronutrients

12.3 Functions, mode of absorption and deficiency symptoms of Micronutrients

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12.4 Deficiency Diseases and symptoms

12.5 Critical Concentration and Toxicity of minerals

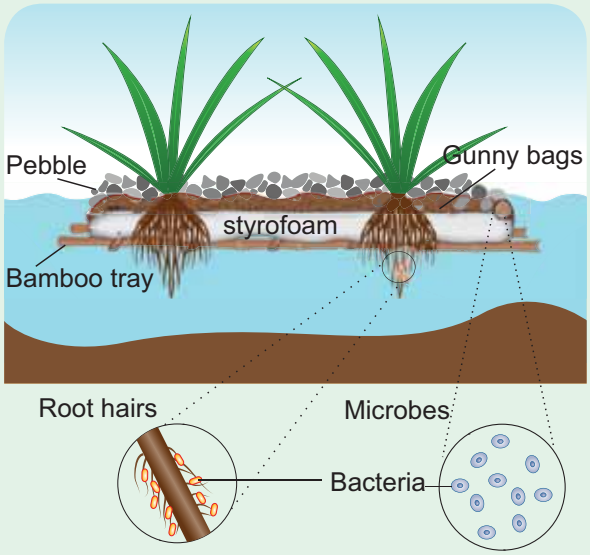
12.6 Hydroponics and Aeroponics

12.7 Nitrogen Fixation

12.8 Nitrogen Cycle and Nitrogen Metabolism

12.9 Special Modes of Nutrition

A solution to Pollution



A new solution has come up for high nutrient pollution and eutrophication in surface waters. Floating Treatment Wetlands (FTWs) offer promising solution and it is a built structure which measures around 3,000 sq.ft and comprises four layers: floatable bamboo at base, styrofoam second layer, a third layer of gunny bags with gravels and final layer to support cleaning agents (plants). Native plants including Vetivers, *Citronella*, Tulsi and *Withania* are being researched for use as cleaning agents. FTW works on the principle of Hydroponics which is explained in this chapter. Microbes grown on the roots of these plants break down and consume organic matter in water and reduce pollution.